Veltra Test Systems

VT-LD20

Pressure-decay leak tester

Operating and Maintenance Manual

Revision 3

For qualified maintenance personnel

1 Safety

1.1 General safety

Only qualified maintenance personnel may service the VT-LD20. Wear safety glasses when you work near the test fixture, because the fixture is pressurised during a test. Never open the fixture while the tester is filling or a test is running.

1.2 Lockout and tagout

Close the compressed air supply and fit your personal lock and tag on the tester pneumatic valve. Compressed air stays in the lines and in the fixture after the supply is closed, so vent the lines and check that the gauge reads zero before you open any fitting. Verify zero energy before you start.

2 Alarm table

The control shows these codes. Likely causes are listed in order of likelihood.

Code	Message	Severity	Likely causes	Action
VT-012	Reference pressure out of tolerance	warning	pressure sensor drift (most likely)	See 4.1 Recalibrate pressure sensor (plant ref PROC-LT-01)
VT-018	Leak rate above limit on master part	critical	worn fixture seals (most likely) pressure sensor drift (possible)	See 4.2 Replace fixture seals (plant ref PROC-LT-02) See 4.1 Recalibrate pressure sensor (plant ref PROC-LT-01)
VT-021	Fill time exceeded	warning	worn fixture seals (most likely)	See 4.2 Replace fixture seals (plant ref PROC-LT-02)

3 Troubleshooting table

Find the symptom, then follow the remedy.

Symptom	Probable cause	Remedy
good parts failing leak test, reference off	pressure sensor drift	See 4.1 Recalibrate pressure sensor (plant ref PROC-LT-01)
leak rate high on every part, fixture hissing	worn fixture seals	See 4.2 Replace fixture seals (plant ref PROC-LT-02)

4 Procedures

4.1 Recalibrate pressure sensor

Plant ref: PROC-LT-01 | Lockout required: no | Time: 30 min

No isolation needed. Work from the HMI only.

1. Load the master part and start Calibration, then Reference, from the tester menu.
2. Compare the reading with the reference gauge.
3. Apply the offset if the difference is under 2 percent, otherwise replace the sensor with SENS-PRS-10.
4. Run three master parts and confirm they pass.

Parts required: none.

4.2 Replace fixture seals

Plant ref: PROC-LT-02 | Lockout required: yes | Time: 45 min

WARNING: Apply lockout and tagout before starting. Verify zero energy.

1. Close the air supply and apply lockout and tagout on the tester pneumatic valve.
2. Remove the fixture plate and both test seals.
3. Fit two SEAL-FIX-30 seals and refit the plate.
4. Remove the lockout and run three master parts to confirm the leak rate is within limits.

Order no.	Description	Qty
SEAL-FIX-30	Leak test fixture seal 30 mm	2

5 Spare parts list

Order spare parts by part number.

Part number	Description	Fits
FLT-AIR-05	Compressed air filter 5 micron	Fill valve
O-RING-NBR-30	O-ring NBR 30 mm, pack of 10	Test fixture and seals
SEAL-FIX-30	Leak test fixture seal 30 mm	Test fixture and seals
SENS-PRS-10	Pressure sensor 0 to 10 bar	Pressure sensor
VALVE-FILL-08	Fill valve 8 mm	Fill valve